

popquiz 19 stellar parameters 3

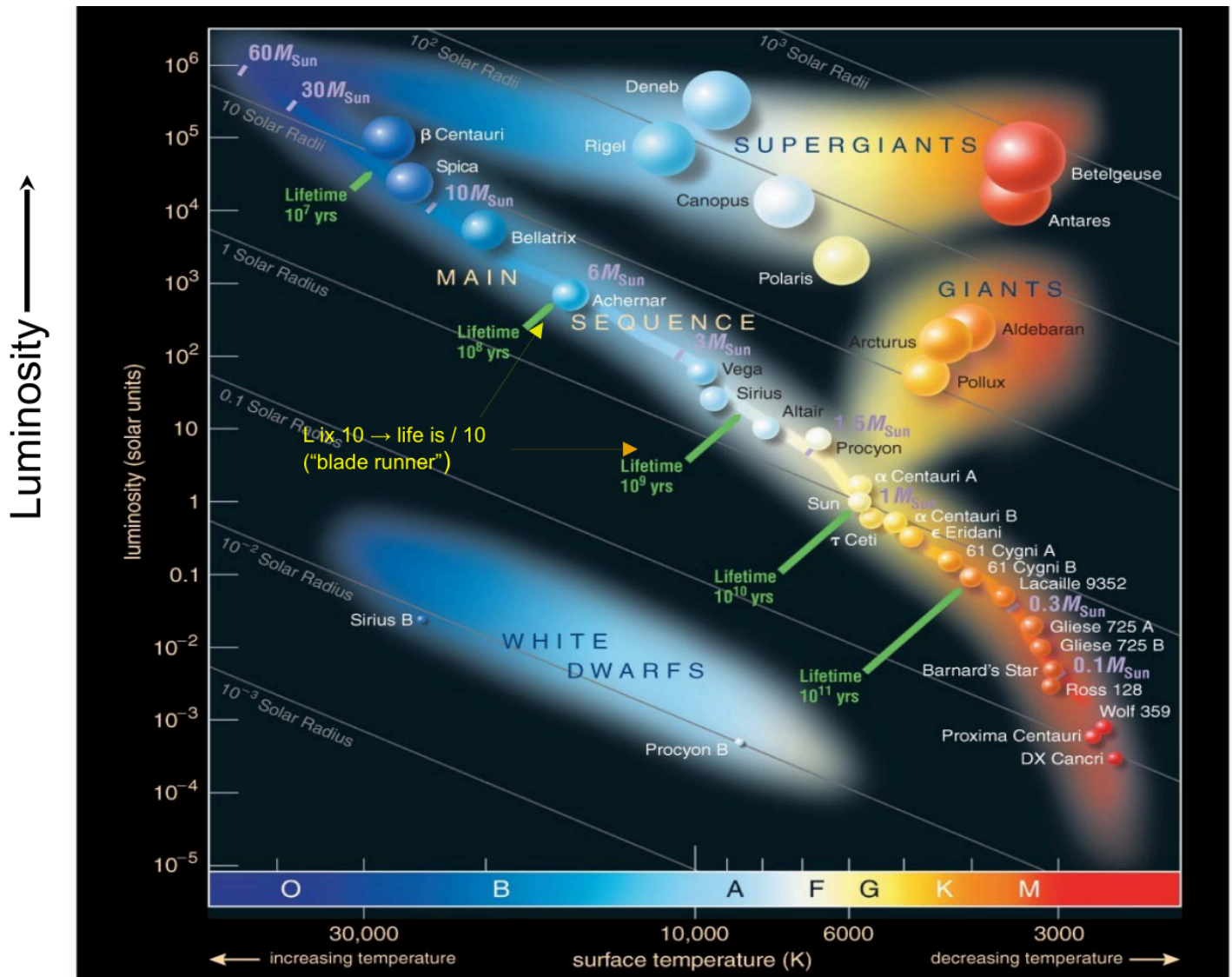
⚠ This is a preview of the draft version of the quiz

Started: Mar 23 at 7:02pm

Quiz Instructions



Question 1 10 pts



On the y-axis, you have luminosity in Sun's luminosity. so 10 means 10 times more luminous than the Sun. 10^{-2} means 100 times less luminous than our sun. 10^2 means 100. 10^3 means 1000. 10^4 means 1000 etc...

Rigel is about 1

☐

as luminous as the Sun

☐

100,000 times more luminous than the Sun

☐

1000 times less luminous than our Sun

☐

1,000 times more luminou than the Sun



Question 2 10 pts

Building on the previous question. Sirius is

☐

a A star

☐

a M star

☐

a G star

☐

a O star



Question 3 5 pts

Building on the previous question.

the red supergiant betelgeuse is

☐

less luminous and cooler than sirius

☐

hotter than sirius but less luminous

☐

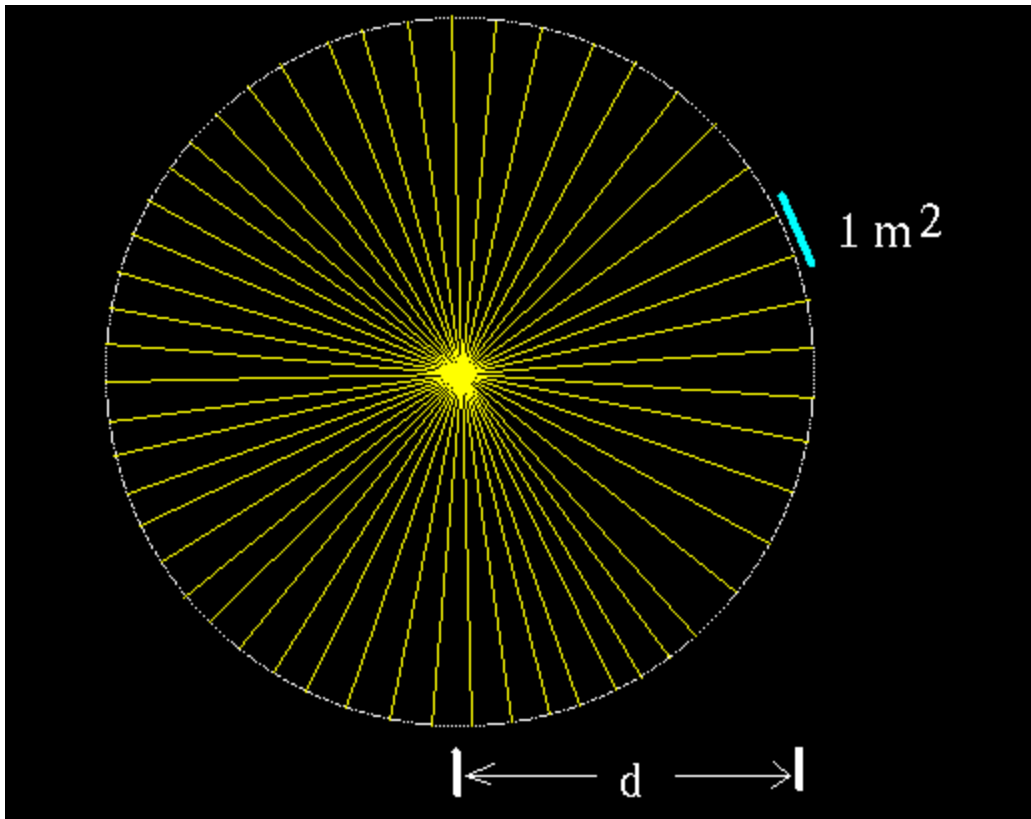
more luminous than sirius but cooler

☐

more luminous than sirius and hotter



Question 4 10 pts



At the center you have a star. Its power output is measured in Watts (energy produced per second). Its in 3D. radiative energy is flowing from the center. At a distance d from the center the amount of aWatts per square meter (per unit area) will be less and is called

(see previous lecture)

- ☐ luminosity
- ☐ relative magnitude
- ☐ brightness
- ☐ absolute magnitude



Question 5 10 pts

A star like Betelgeuse has a temperature smaller than the Sun's. But it is 100,000 (about) more luminous. This is because

hint: in the HR diagram its in the upper right corner.

- ☐ it is blue when the Sun is white

☐

it is larger

☐

it burns hydrogen in its core faster

☐

it spins faster



Question 6 10 pts

unit 7.3 slide 6

Say 2 stars A and B have the same temperature. But Star A is 1000 larger. That means

(Area is proportional to size²)

☐

star A is 1 million times more luminous

☐

star B is 1 million times more luminous

☐

star B is 1000 times more luminous

☐

star A is 1000 times more luminous



Question 7 10 pts

unit 7.2 .

Consider a 100 W light bulb and you are close to it . 100 W is its luminosity. Power output. If you move away at a distance 10 times larger, the luminosity (brightness) is now

☐

100 W (per m²)

☐

10 W (per m²)

☐

1000 W (per m²)

☐

1 W (per m²)



Question 8 10 pts

The Sun is a powerful "light bulb" (4×10^{26} W). This energy spread in 3D in space. At a distance 93 million miles from the Sun, the brightness (power per unit area) is?

hint: which planet is at this distance? see unit 7.2 .



4 10^6 per square meter



1000 W per square meter



100W per square meter



100,000 W per square meter



Question 9 10 pts

Luminosity is to brightness what absolute magnitude is to



apparent brightness



relative magnitude



declination



apparent magnitude



Question 10 10 pts

Consider the HR diagram. On the y-axis you have



temperature



spectral classes



luminosity



distance from earth



Question 11 10 pts

The Sun has a life of 10 billion years (10^{10} or 10,000,000,000) . A star 10 times more luminous has a life of

(remember the quote from blade runner. unit 7.1 slide 5)



1 trillion years

☐

100 million years

☐

1 billion years

☐

1 million years



Question 12 10 pts

Building on the same question.

If a star is 10 times less luminous than the Sun. Its life is

☐

100 billion years

☐

100 million years

☐

10 trillion years

☐

10 billion years

Not saved

Submit Quiz